COLLATERALIZED DEBT OBLIGATIONS

Learning Outcome

- describe collateralized debt obligations, including their cash flows and risks

Collateralized debt obligations (CDOs) issue securities backed by a diversified pool of one or more debt obligations. CDOs can be backed by a broad range of debt. CDOs backed by corporate and emerging market bonds are **collateralized bond obligations (CBOs)**; those backed by leveraged bank loans are **collateralized loan obligations (CLOs)**; those backed by other CDOs are structured finance CDOs; and those backed by a portfolio of credit default swaps for other structured securities are synthetic CDOs. The prevailing CDO structure is the CLO structure, where the collateral pool is made up of leveraged bank loans.

Exhibit 7 summarizes features of CDOs, covered bonds, mortgage-backed securities (MBS), and non-mortgage ABS:

Exhibit 7:

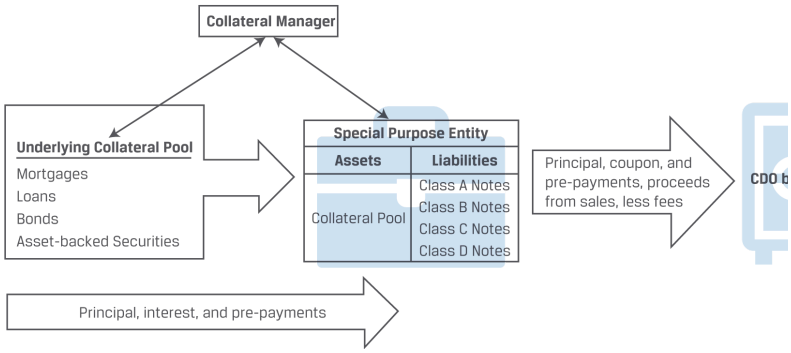
Overview of CDOs and Other Securitized Products

	Covered Bonds	CDO	MBS	Non-Mortgage ABS
Collateral	Commercial or residential mortgages, or public sector assets	Leveraged bank loans (CLOs)	Commercial and residential mortgage loans	Credit card receivables (non-amortizing) and Solar lease/loan payments
Impact on Issuer Balance Sheet	Collateral remains on the balance sheet and ringfenced into a separate cover pool	Collateral removed from balance sheet	Collateral removed from balance sheet	Collateral removed from balance sheet
Number of Bond Classes	One bond class with its associated default exposure in its cover pool	Typically several	Typically several	Typically several
Collateral Pool	Unstable with ongoing collateral management	Unstable with ongoing collateral management	Stable	Unstable with ongoing collateral management; a pre-funding period is used by solar ABS post-transaction
Recourse	Dual recourse nature: first on the ringfenced loans in the cover pool and second on the unencumbered assets of the issuing institution	Single recourse nature	Single recourse nature	Single recourse nature

Similar to the ABS structures discussed earlier, the CDO structures redistribute the cash flows the underlying collateral pool generates and redistribute them to the tranches. For the majority of CDOs, the collateral pools are not static, so there is a need for a **collateral manager** that buys and sells debt obligations for and from the CDO's collateral pool to generate sufficient cash flows to meet the obligations to the CDO bondholders. Otherwise,

Exhibit 8:

The Structure of a Generic CDO Transaction



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The proceeds to pay the CDO bond classes can come from interest payments from collateral assets, maturing of collateral assets, and sale of collateral assets. The basic economics of these CDO transactions is to ensure that the return on the collateral pool is higher than the funding costs, the aggregate cost of the bond classes issued to finance the transaction. Holders of senior and mezzanine bonds earn fixed returns, while the equity tranche holders and the CDO manager earn more equity-like returns. Essentially, a CDO is a leveraged transaction, where equity tranche holders use borrowed funds (the bond classes issued) to generate a return above the funding cost.

The CDO market has changed significantly since the 2003–2007 period. The collateral today is primarily leveraged bank loans (i.e., CLOs), while during the 2003–2007 period, it consisted of a wide range of products, including different types of real estate debt and commercial mortgage-backed securities (CMBS). While subsequent regulatory and legal changes made arranging and investing in CDOs less attractive, the underlying securitization framework remains similar today. It can be best explored through CLOs, collateralized by senior secured bank loans creating a diversified portfolio of company loans.

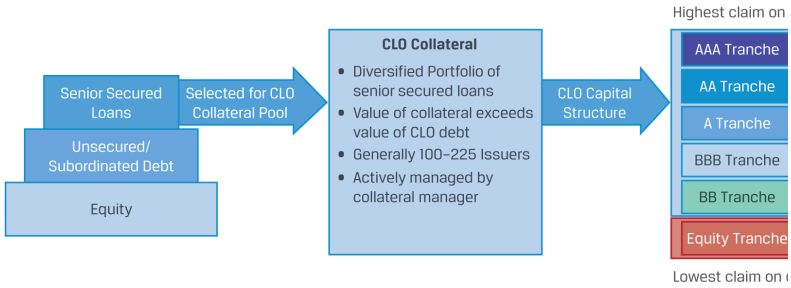
Generic CLO Structure

The capital structure of a generic CLO transaction is shown in Exhibit 9. The funds to purchase the collateral assets for a CLO are obtained from the issuance of debt obligations. These tranches include senior, mezzanine, and subordinated/junior/equity tranches.

Investors in senior or mezzanine bond classes earn a potentially higher yield than comparable corporate bonds offer, or they gain exposure to debt products that they may not otherwise be able to acquire. Investors in equity tranches take on equity-like risks with the potential to earn returns comparable to equities. Moreover, the residual tranche plays a key role in whether a CLO is viable or not; the CLO structure has to offer competitive returns for this tranche.

CLOs come in multiple flavors. These are the most common:

- Cash Flow CLOs, where the cash flows from interest payments and principal repayments are redistributed across the tranches. This is the most common CLO structure.
- Market Value CLOs, where the value accruing to the tranches depends on the market value of the portfolio.
- Synthetic CLOs, where the collateral pool is created synthetically through credit derivatives.



The ability of the manager to make the interest payments and principal repayments depends on collateral performance. The collateral manager must continually meet various performance tests and collateral limits for the underlying collateral. If the manager fails pre-specified tests, a provision is triggered that requires the payoff of the principal to the senior bond class until the tests are met. This process effectively deleverages the CLO because the cheapest funding source for the CLO, the senior bond class, is reduced.

A typical feature of CLO transactions is that the collateral portfolio is not finalized until after the transaction closes. While the collateral manager acquires most of the loans before the transaction closes, there is a subsequent ramp-up period when additional assets are added to the collateral pool. After this period, the manager may replace loans in the portfolio as long as the new asset meets the portfolio selection criteria. The final phase of the CLO lifecycle is when the underlying loans mature and the collateral manager uses the proceeds to pay off the tranches according to the order of payments. Recourse is typically limited to the collateral pool, with minimal recourse to the original issuers.

The cash flow waterfall, in connection with performance-based tests, provides varying degrees of protection to the CLO's debt tranches. The most senior and highest-rated AAA tranche has the lowest yield but enjoys the highest claim on the cash flow distributions and is the most loss-remote. Mezzanine tranches pay higher coupons but are more exposed to loss and have lower ratings. The most junior tranche, equity, is the riskiest, is not rated, and does not have a set coupon. Instead, the equity tranche represents a claim on all excess cash flows once the obligations for each debt tranche have been met.

CLOs face a series of coverage tests to help ensure the cash flows generated by the underlying bank loan collateral meet the distribution obligations in the various CLO tranches. One such test is an overcollateralization test, which helps to keep the principal value of a CLO's underlying bank loan pool above the total principal value of the notes issued by the various CLO debt tranches as long as the CLO debt remains outstanding. If the principal value declines below the overcollateralization test trigger value, cash will be diverted away from equity and junior CLO debt tranches toward senior debt tranche investors.

Consider a hypothetical CLO created with USD700 million of principal promised to the owners of its various debt tranches. To meet this USD700 million obligation, investors and rating agencies may require that the CLO manager use the capital raised from the CLO's debt and equity issuance to purchase USD840 million of bank loans. This would result in an overcollateralization ratio of 1.20.

Each CLO debt tranche has its own targeted overcollateralization ratio. The overcollateralization ratios for each tranche act as covenants and, when tripped, redirect cash flows to purchase additional bank loan collateral or repay the senior-most CLO debt tranche.

CLOs are also subject to a variety of other tests that act in concert in an effort to help protect debt investors from loss. Examples of these tests include a measurement of the industry diversification in the underlying collateral pool of bank loans and the CLO's exposure to non-senior secured loans. Other tests consider the diversity of borrowers

For CLO transactions, the role of the collateral manager is key. The skill the manager displays in selecting, managing, and exchanging assets in the pool directly impacts the performance of the CLO. In a way, the collateral manager acts as an active bond portfolio manager by determining the asset mix and the risk and exposures in the collateral pool.

EXAMPLE 6

CLOs and Systematic Risk

Institutional investors, such as banks, insurance companies, hedge funds, mutual funds, and pension plans, are the primary buyers in the CLO market. Each of these investors targets different CLO tranches. Due to regulations, many banks almost exclusively buy the AAA-rated tranche. Other investors, such as hedge funds and endowments, may be more speculative and thus take more risk to earn higher returns. Current CLO structures are typically simpler than the CDO structures that played a pivotal role in the GFC. Structured finance CDOs, which are made up of other CDOs, are no longer common, and even synthetic CLOs that use derivatives are less prevalent.

After the GFC, the underlying asset quality of CLOs has received more scrutiny than was previously common. This is a reflection of the need to understand the unique characteristics of the complex asset pool in a CLO.

Question Set

1. Describe how a CDO is distinguished from a CMO by the function of the CDO's collateral manager.

Hide Solution

Solution:

While CDOs are similar to CMOs in that both issue securities backed by a diversified pool of one or more debt obligations, CDO collateral pools usually are not static. Their volatility generates a need for a collateral managerthat buys and sells debt obligations for and from the CDO's collateral pool to generate sufficient cash flows to meet the obligations to the CDO bondholders.

2. Identify the CLO type that corresponds to its correct characteristic:

i. Relies on OTC contracts
ii. Most common CLO structure
iii. Sizes of tranches have greatest variability

A. Cash flow CLO	
B. Market value CLO	
C. Synthetic CLO	

Hide Solution

Solution:

A. Cash flow CLO	ii. Most common CLO structure
B. Market value CLO	iii. Sizes of tranches have greatest variability
C. Synthetic CLO	i. Relies on OTC contracts

with certainty.

C. i is correct. The collateral pool of synthetic CLOs are created synthetically through credit derivatives, which are financial contracts traded OTC.

3. Explain why a CDO's equity tranche holders are *most likely* to earn the highest returns when the CDO works as designed.

Hide Solution

Solution:

The basic economics of CDO transactions is to ensure the return on the collateral pool is higher than the funding costs, the aggregate cost of the bond classes issued to finance the transaction. Holders of senior and mezzanine tranches earn fixed returns, while the equity tranche holders and the CDO manager earn more equity-like returns. The equity tranche represents a claim on all excess cash flows once the obligations for each debt tranche have been met, so it is the riskiest tranche and should earn the highest return. Essentially, a CDO is a leveraged transaction, where equity tranche holders use borrowed funds (the bond classes issued) to generate a return above the funding cost.

4. Describe the effect of collateral performance test failure on a CLO's financial structure.

Hide Solution

Solution:

In making interest and principal payments, a CLO collateral manager must continually meet various performance tests and collateral limits for the underlying collateral. Upon a test failure, a provision is triggered requiring payoff of the principal to the senior bond class until the tests are met. This effectively deleverages the CLO because the cheapest funding source for the CLO, the senior bond class, is reduced.

5. In certain CLOs, the motivation is to capture a spread between the potential collateral return and the funding cost. An additional risk of this type of securitization relative to an investment in an asset-backed security is:

A. the default risk on the collateral assets.
B. the risk that the CLO manager will fail to earn a return sufficient to pay off the investors in the senior and the mezzanine tranches.
C. the risk due to the mismatch between the collateral making fixed-rate payments and the bond classes making floating-rate payments.

Hide Solution

Solution:

B is correct. In addition to the risks associated with investments in ABS, such as the default risk on the collateral assets (answer A) and the risk due to the potential mismatch between the collateral making fixed-rate payments and the bond classes making floating-rate payments (answer C), investors in CLOs face the risk that the CLO manager will fail to earn a return sufficient to pay off the investors in the senior and the mezzanine tranches. With an ABS, the cash flows from the collateral are used to pay off the holders of the bond classes without the active management of the collateral—that is, without a manager altering the composition of the debt obligations in the pool that is backing the securitization. In contrast, in a CLO, the manager buys and sells debt obligations with the dual purpose of not only paying off the holders of the bond classes but also generating an attractive/competitive return for the equity tranche and for the manager.

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issuers.

B. Investors in senior or mezzanine bond classes typically earn a lower yield than comparable corporate bonds.

C. After the ramp-up period, the collateral manager may replace loans in the portfolio as long as the new asset meets the portfolio selection criteria.

Hide Solution

Solution:

The correct answer is B. Investors in senior or mezzanine bond classes earn a potentially *higher* yield than comparable corporate bonds offer. A is incorrect because it is a correct statement. Recourse is limited to the collateral pool, with minimal recourse to the original issuers. C is incorrect because it is a correct statement. The collateral manager acquires most of the loans before the transaction closes. In the subsequent ramp-up period, additional assets are added to the collateral pool. After this ramp-up period, the manager may replace loans in the portfolio as long as the new asset meets the portfolio selection criteria.

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The Bloomberg SECF function allows you to view a list of asset-backed securities.

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	SSA	CF	Ticker	Series	Class	Cpn	Maturity	Orig Amt	Factor	Pvt	Type	Ctry/...	Curr	Issued
1)		PD	AAA	2003-1A	A1	1.580	09/27/33	270,000,000	0.00000000	Y	SC,FLT...	US	USD	06/16/03
2)		PD	AAA	2003-1A	A2	2.034	09/27/33	44,100,000	0.00000000	N	SC,AFC...	US	USD	05/01/03
3)		PD	AAA	2003-1A	A3	2.505	09/27/33	88,000,000	0.00000000	N	SC,AFC...	US	USD	05/01/03
4)		PD	AAA	2003-1A	A4	3.454	09/27/33	51,700,000	0.00000000	N	SC,AFC...	US	USD	05/01/03
5)		PD	AAA	2003-1A	A5	3.750	09/27/33	46,783,166	0.00000000	Y	SC,AFC...	US	USD	05/01/03
6)		PD	AAA	2003-1A	A6	3.750	09/27/33	41,000,000	0.00000000	Y	SC,AFC...	US	USD	05/01/03
7)		PD	AAA	2003-1A	X	8.586	09/27/33	541,583,166	0.00000000	Y	SC,IO,...	US	USD	05/01/03
8)		PD	AAA	2003-1A	R	0.000	09/27/33	0	0.00000000	Y	SC,R	US	USD	05/01/03
9)		PD	AAA	2005-1A	1A1A	5.485	02/27/35	140,000,000	0.00000000	Y	SC,FLT	US	USD	02/22/05
10)		PD	AAA	2005-1A	1A1B	5.485	02/27/35	156,100,000	0.00000000	Y	SC,FLT	US	USD	02/22/05
11)		PD	AAA	2005-1A	1A2	5.590	02/27/35	227,480,000	0.00000000	Y	SC,FLT	US	USD	02/22/05
12)		Y	AAA	2005-1A	1A3A	5.540	02/27/35	45,000,000	0.04428600	Y	SC,FLT	US	USD	02/22/05
13)		Y	AAA	2005-1A	1A3B	5.570	02/27/35	84,694,029	0.04428600	Y	SC,FLT	US	USD	02/22/05
14)		Y	AAA	2005-1A	1X	0.000	02/27/35	653,274,029	0.00879200	Y	SC,IO,...	US	USD	01/01/05
15)		PD	AAA	2005-1A	2A1	2.763	02/27/35	144,200,000	0.00000000	Y	SC,FLT	US	USD	02/22/05
16)		Y	AAA	2005-1A	2A2	5.590	02/27/35	80,294,000	0.11709500	Y	SC,FLT	US	USD	02/22/05
17)		Y	AAA	2005-1A	2X	0.000	02/27/35	224,494,000	0.04188100	Y	SC,IO,...	US	USD	01/01/05
18)			AAA	2005-1A	R	0.000	02/27/35	0	0.00000000	Y	SC,R	US	USD	01/01/05

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